Continuous System Simulation Languages (CSSLs)

Exam-Oriented Notes | Simulation and Modelling | Chapter 3.8

1. Definition

Continuous System Simulation Languages (CSSLs) are simulation languages that use familiar statement-type input for digital computers. A problem can be programmed **directly from the equations of a mathematical model**, so the equations do not need to be broken into functional elements (as is required on an analog computer).

2. Key Features

- Provide simple analog-type functions such as **addition and integration**.
- Contain **algebraic and logical expressions** to describe relationships between variables.
- Easy to learn and apply to problems in all sciences and engineering disciplines.
- Problems can be **coded in a short time, executed immediately, and evaluated quickly** by inspecting graphic output in several forms.
- Used together with software tools that solve the **differential equations** of the system under study.

3. Purpose and Applications

- Help engineers and scientists to **model, analyze and evaluate** the dynamic behavior of physical phenomena.
- Simulate system behavior under **different conditions and inputs**.
- **Optimization of system design.**
- **Prediction of system performance.**
- **Identification of potential problems or failures.**

4. Example: CSMP III

CSMP III (Continuous System Modeling Program, Version III) is an early computer software package designed for modelling and solving differential equations numerically.

Three Types of Statements in a CSMP III Program

Statement Type	Description
Structural statements	FORTRAN-like statements and functional blocks designed for repeated operations that frequently occur in the model definition.
Data statements	Assign numerical values to **parameters, constants and initial conditions**.
Control statements	Specify options in the **assembly and execution** of a program and the **choice of output results**.

Equation Conversion Example

Mathematical model:

$$Y = 6X/W + (Z - 3)^2$$

Equivalent CSMP III statement:

```
Y = 6.0*X/W + (Z - 3.0)**2.0
```

Note: * means multiplication and ** means power (exponent).

5. Some CSMP III Functional Blocks

S.N.	Function	Mathematical Form	General Form in CSMP III
1	**Integrator**	Y = integral of X dt from 0 to t, plus IC	`Y = INTGRL(IC, X)` with Y(0) = IC
2	**Exponential**	$Y = e^X$	`Y = EXP(X)`
3	**Natural logarithm**	Y = ln(X)	`Y = ALOG(X)`
4	**Trigonometric sine**	Y = sin(X)	`Y = SIN(X)`

*(The textbook writes "natural algorithm"; the correct term is natural **logarithm**. IC = Initial Condition.)*

6. Quick Revision Points

- CSSL = equation-oriented programming of continuous models.
- No need to break equations into functional blocks.
- CSMP III has **3 statement types**: Structural, Data, Control.
- **INTGRL(IC, X)** is the integrator block, where IC = initial condition.
- Output is mainly **graphical**, allowing quick evaluation of results.

7. Likely Exam Questions

S.N.	Question	Marks
1	What is a CSSL? Explain its features and uses.	5
2	Explain CSMP III and its three types of statements.	5
3	Write the CSMP III statement for $Y = 6X/W + (Z - 3)^2$.	2
4	List CSMP III functional blocks with their general forms.	3